Stochastic processes and stationarity

WHAT CAN BE FORECASTED - RECAP: 1/2

- there is some *structure* in the timeseries
- sufficient historical data
- stationarity (future is similar to the past).

In many cases, **these condition holds**. If they don't, we can try to force them:

- remove cycles and trends
- remove effect of the forecast on the signal
- transform the data

We have some tools that help understanding whether a given time series is forecastable. These tools are based on a stochastic process representation of time series.

One counterexample:



Counterexample: CISCO stock prices between 2000 and 2002

Forecasting

Forecasting is about predicting the future as **accurately** as possible, given all available information, including historical data and knowledge of future events that could affect forecasts.

$\hat{x}_{t+h|t}$

Time-series forecasting is identifying time-invariant relations and components such as **patterns** and *(auto)(cor)*relations in the data.

Dual definition

Forecasting is the process of quantifying the uncertainty carried by the future. Indeed forecasting can be described as the art of making ***residuals stationary*** and unpredictable.

A time series is a collection of values of a variable x over time:

$$\{x_t; t \in \mathcal{T}\}$$

where $\mathcal{T}$ can be regularly or irregularly spaced. Most of the literature assumes **discrete regularly sampled** time series. We can think at this collection as realizations of stochastic variables.

Stochastic processes

A **probability space** is a triple $(\Omega, \mathcal{A}, P)$, where

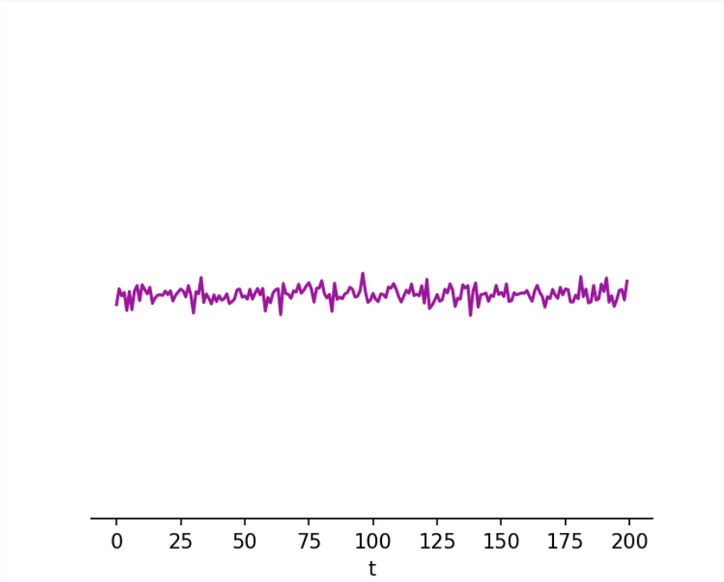
1. Ω is a nonempty set, we call it the sample space.
2. $\mathcal{A}$ is a σ-algebra of subsets of Ω, i.e. a family of subsets closed with respect to countable union and complement with respect to Ω.
3. P is a probability measure defined for all members of $\mathcal{A}$ That is, a function $P : \mathcal{A} \to [0, 1]$ such that $P(\mathcal{A}) \geq 0$ for all $A \in \mathcal{A}$, $P(\Omega) = 1$, $P\left(\cup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P\left(A_i\right)$ for all sequences $A_i \in \mathcal{A}$

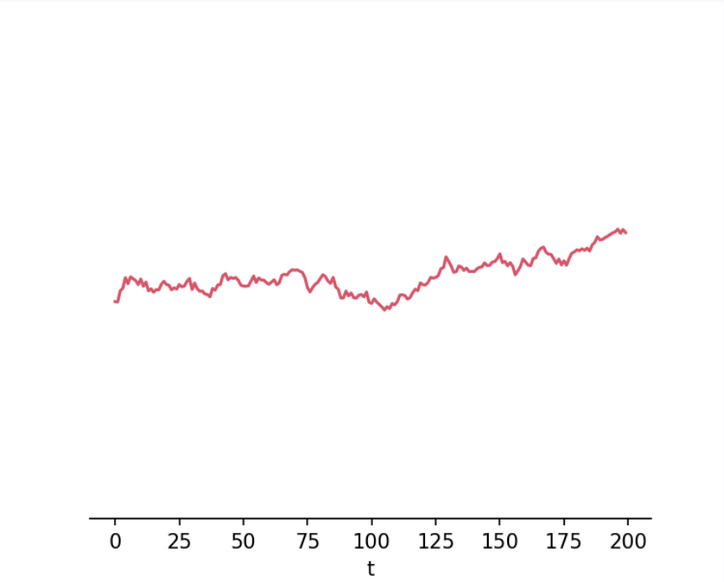
A **stochastic process** is a family of random variables defined on the same probability space $(\Omega, \mathcal{A}, P)$:

$$\{x_t(\omega); t \in \mathcal{T}\}$$

Example of stochastic processes

$x_t(\omega) \sim N(0,1)$ $\quad$ $x_t(\omega) \perp x_s(\omega)$ for $t \neq s$ $\quad$ $x_t(\omega) - x_{t-1}(\omega) \sim N(0,1)$





Joint distribution - 1

The joint distribution function of $(x_{t_1}(\omega), x_{t_2}(\omega), \ldots, x_{t_s}(\omega))$ can be defined as:

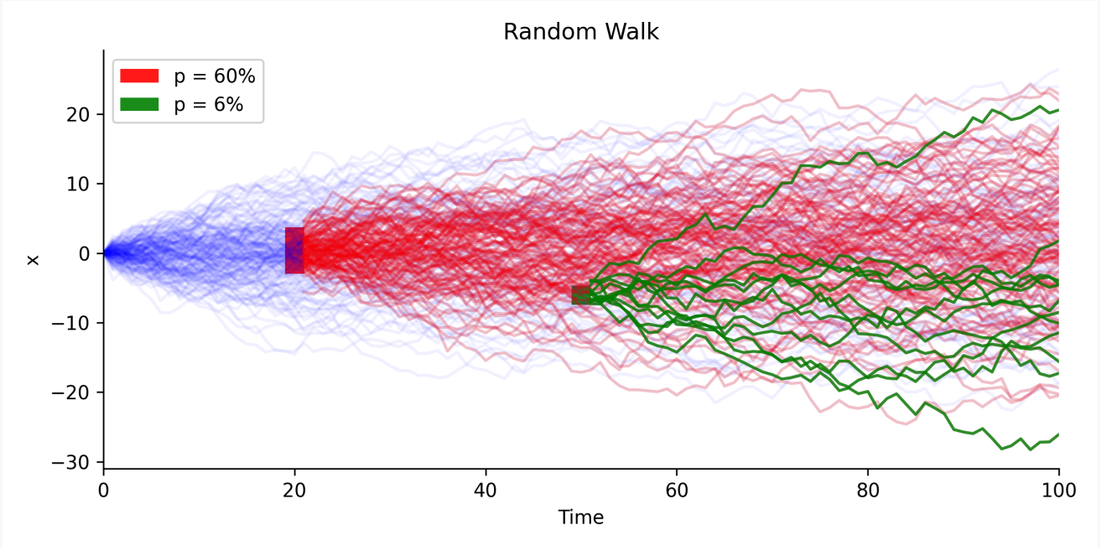
$$F_{t_1,t_2,\cdots,t_s}(b_1, b_2, \cdots, b_s) = \mathrm{P}\left(x_{t_1}(\omega) \leq b_1, x_{t_2}(\omega) \leq b_2, \ldots, x_{t_s}(\omega) \leq b_s\right))$$

The above expression is known also as a **copula**, that is, a cumulative density function over a **multivariate space**.

Joint distribution - 2

If F was known for a particular stochastic process $x_t(\omega)$, we could answer questions such as:

1. Which is the probability that the process $\{x_t(\omega); t \in \mathbb{Z}\}$ passes through [a, b] at time t?
2. Which is the probability that the process $\{x_t(\omega); t \in \mathbb{Z}\}$ passes through [a, b] at time t_1 and through [c, d] at time t_2?

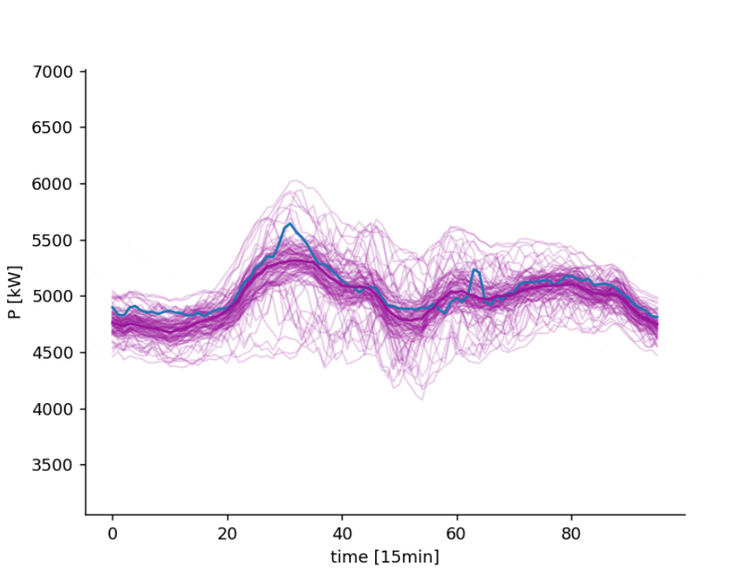


Joint distribution - 3

It turns out that this ability is pretty much all we need to know about the stochastic process.

> **Theorem** Under general conditions the probabilistic structure of the stochastic process $\{x_t(\omega); t \in \mathbb{Z}\}$ is completely specified if we are given the joint probability distribution of $(x_{t_1}(\omega), x_{t_2}(\omega), \ldots, x_{t_s}(\omega))$ for all values of s and for all choices of $\{t_1, t_2, \cdots, t_s\}$
> *(Priestly 1981, p.104)*

In fact, once F is known, we can generate an **infinite** set of realizations from the stochastic process
$x_t(\omega_i) \quad i = [1, 2, ...\infty]$, also called **scenarios**



7 days ahead forecast of the power consumption of a distribution system operator. Blue: observed. Violet: point forecast and possible scenarios

Joint distribution - 4

If the future is not similar to the past, we cannot estimate $F_{t_1,t_2,\cdots,t_s}$ for arbitrary values of s. Even for a univariate stochastic process $\{x_t(\omega); t \in \mathbb{Z}\}$, where $x_t \sim N\left(\mu_t, \sigma_t^2\right)$, this requires to estimate

$$F_t(b) = \int_{-\infty}^{b} \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left\{-\left(\frac{v-\mu_t}{\sigma_t}\right)^2\right\} dv \text{ for } t = 0 \pm 1, \ldots t$$

To reduce the number of parameters to be learned (in the example above $\{(\mu_t, \sigma_t)_{t=1}^{t_s}\}$), some regularity conditions must hold.

- the TS must be somehow homogeneous in time
- the process must have limited memory

Let's do a step back and try to ask a somehow simpler question:

> Is it possible to estimate statistical moments of the time series?

Since we can only observe one realization of the stochastic process in time, **estimates for mean and variance can only be done by time averaging** (as opposed to sample path average).

Stationarity

A process is said to be stationary (in a given window) if sliding along the time axis the joint CDF doesn't change

$$F_{t_1+k,t_2+k\ldots t_n+k} = F_{t_1,t_2\ldots t_n}$$

- Many forecasting techniques assume it or work better in its presence
- The raw data is rarely stationary, we should check stationarity **after accounting for patterns**

Ergodicity

Additionally, a process is said to be **ergodic** w.r.t. one of its moment if **we can estimate them by observing a sufficiently long realization of the process** and if estimations converge quadratically to the true values.

In other words, imagine to observe a set of sample paths $x_j = \{x_{j,t_1}, x_{j,t_2}, x_{j,t_3} \ldots x_T\}$. The process is ergodic if averaging over different x_j gets the same results of averaging over time.

? Which one of the two examples of the two previous stochastic processes is ergodic?

Ergodicity - Slutsky's Theorem

Let x_t be a stationary process with mean μ and autocovariance γ_k If: $lim_{k\to\infty}\gamma_k = 0$ or $\sum_{k=0}^{\infty}|\gamma_k| < \infty$ then x_t is mean-ergodic

Intuitively, this result tell us that if any two random variables positioned far apart in the sequence are almost uncorrelated, then some new information can be continually added so that the sample mean will approach the population mean.

By the **law of large numbers**, good estimates of mean and covariance for ergodic processes are:

$$\hat{\mu}_T(t) = \frac{1}{T}\sum_t^{t+T} x_t$$

$$\hat{\gamma}_x(k) = \frac{1}{T}\sum_t^{t+T} (x_t - \hat{\mu}_T)(x_{t-k} - \hat{\mu}_T)$$

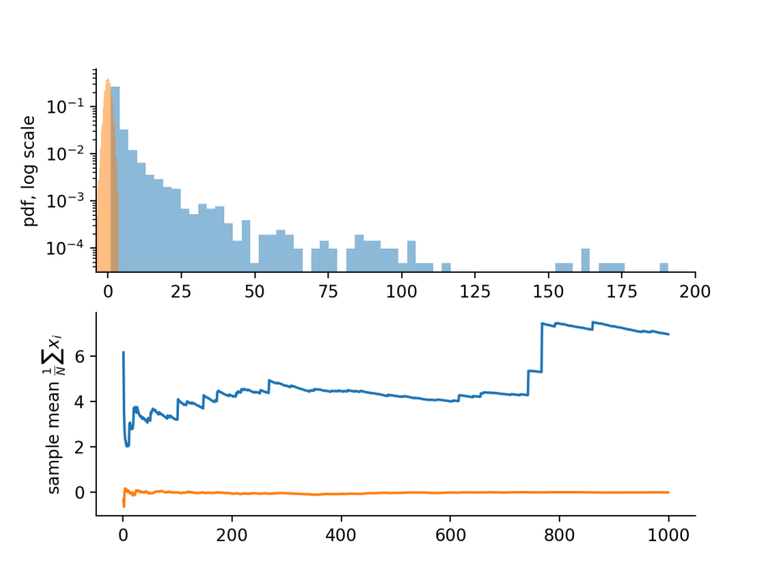
Caveat - Fat tail distributions

Beware! In some cases these estimations do not converge quickly

Consider the case in which the stochastic process has distribution:

$$F_X(x) = \begin{cases} 1 - \left(\frac{x_\mathrm{m}}{x}\right)^\alpha & x \geq x_\mathrm{m} \\ 0 & x < x_\mathrm{m} \end{cases}$$

This is known as Pareto distribution, which is a power law distribution. These are also known as fat tail distributions. Two notable examples of **fat-tailed** natural variables are **earthquakes** and **pandemics**.



Top: pdfs for normal and pareto distribution with alpha = 1.12. Bottom: convergence of sample means

ACF function

The autocorrelation function is defined as:

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{cov(x_t, x_{t-k})}{\sigma(x_t)\sigma(x_{t-k})} \overset{stationary}{=} \frac{cov(x_t, x_{t-k})}{\sigma^2(x)} = \frac{\sum_{t=1}^{n-k}(x_t - \bar{x})(x_{t+k} - \bar{x})}{\sum_{t=1}^{n}(x_t - \bar{x})^2}$$

- **Before** computing **ACF** is usually a good idea to try to **make the time-series stationary**, e.g. by removing a linear trend.
- If a series has a trend (it is non-stationary), the ACF will show strong positive correlations for many lags simply because "high values tend to follow high values."

ACF as an IID test

The ACF function can be used to construct the following hypothesis test:

$H_0 : x_t \sim i.i.d.(0, \sigma^2)$

if a process is IID, we have that $\rho_k = 0$.

A decision rule for the hypothesis test could be:

reject H_0 if $|\hat{\rho}_k| > c$

that is, for a given lag k we require the magnitude of the correlation to be less than a constant value.

ACF as an IID test - 2

How do we choose c? We can request a 95% confidence level.

$$\mathbb{P}(|\hat{\rho}_k| < c|H_0) = 0.95 \mathbb{P}(-c \leq \hat{\rho}_k \leq c|H_0) = 0.95$$

We know that for white noise $\hat{\rho}_k \sim \mathcal{N}(0, 1/n)$

This means that $\mathbb{P}(-c \leq \hat{\rho}_k \leq c|H_0) = 0.95$ iff $c = \frac{1.96}{\sqrt{n}}$

Demonstration

$$\rho_k = \frac{\sum_{t=1}^{n-k} x_t x_{t+k}}{\sum_{t=1}^{n} x_t^2}$$

1. $\mathbb{E}\rho_k = 0$: since $x_t \perp x_{t+k}$, the expectation of the numerator is 0
2. $\mathbb{V}\rho_k = 1/n$. Since
3. $\mathbb{V}(x_t x_{t+k}) = \mathbb{E}(x_t^2 x_{t+k}^2) - [\mathbb{E}(x_t x_{t+k})]^2 = \mathbb{E}(x_t^2)\mathbb{E}(x_{t+k}^2) = \sigma^2\sigma^2 = \sigma^4$
4. $\mathbb{V}\rho_k = \frac{(n-k)\sigma^4}{(n\sigma^2)^2} \approx \frac{1}{n}$
5. For the central limit theorem, since ρ_k is estimated summing iid random variables $x_t x_{t+k}$, the distribution of its plug in estimator is normally distributed $\mathcal{N}(0, \mathbb{V}[\rho]))$

PACF

The partial auto-correlation function is defined as the ACF of the residual between the time series and the best linear fit with its first k lags

$$X_t = \phi_{k1} X_{t-1} + \phi_{k2} X_{t-2} + \cdots + \phi_{kk} X_{t-k} + \varepsilon_t$$

This function is particularly useful for identifying the order of an autoregressive (AR) process. For example, if the PACF cuts off after a certain lag, it suggests that an AR model of that order might be appropriate.

Portmanteau tests

When we look at the ACF plot to see whether each spike is within the required limits, we are implicitly carrying out multiple hypothesis tests, each one with a small probability of giving a false positive. When enough of these tests are done, it is likely that at least one will give a false positive, and so we may conclude that the residuals have some remaining autocorrelation, when in fact they do not. Instead of testing a given lag k, we can test for a set of lags. For an i.i.d. process, the following random variables are approximately chi-square with K degrees of freedom.

$$Q_K = T\sum_{k=1}^{K}\hat{\rho}_k^2$$

Box-Pierce test

$$Q_K^* = T(T+2)\sum_{k=1}^{K}(T-k)^{-1}\hat{\rho}_k^2$$

Ljung-Box test

For these tests, the null hypothesis is that the time series is white noise. If we get a small p-value from the test, we can reject the null hypothesis

Since stationarity is required to estimate the statistical properties of a time series, many statistical models require this condition to make effective and precise predictions.

Augmented Dickey Fuller ("ADF")

H_0 The series is non-stationary

We look for evidence that the null hypothesis is false. Consequently, small p-values (e.g., less than 0.05) rejecting H_0 means that we can consider the series stationary.

Kwiatkowski-Phillips-Schmidt-Shin ("KPSS")

H_0 The series is trend-stationary (has no unit root)

We look for evidence that the null hypothesis is false. Consequently, small p-values (e.g., less than 0.05) rejecting H_0 means that we can consider the series non stationary.

Some forecasting methods are extremely simple and yet surprisingly effective. They are stateless methods

Naïve method

This method performs well for $h = 1$, or at scale which are smaller than the characteristic time scale of the signal

$$\hat{y}_{t+h|t} = y_t$$

Mean method

We just take the mean value of the time series observed so far

$$\hat{y}_{t+h|t} = \bar{y} = (y_1 + \cdots + y_t)/t$$

Conditional Mean method

We take the mean of the observations that were registered in similar conditions

$$\hat{y}_{t+h|t} = \frac{1}{T} \sum_{t \in \mathcal{T}_\S} y_t \quad \mathcal{T}_\S = \{t : dist(x_t, x_{t+h}) < \epsilon\}$$

Example: I'm predicting sales for a given product on an online platform. The platform applies time-varying discounts based on different market conditions. We can forecast the product taking the mean sales in the same day of the week when the same discount was applied.

Seasonal Naïve method

$$\hat{y}_{t+h|t} = y_{t+h-m(k+1)}$$

where m is the period of the seasonality and $k = \lfloor (h-1)/m \rfloor$

Assuming residuals are normal, uncorrelated:

Mean: $y_{T+h|T} \sim N(\bar{y}, (1 + 1/T)\hat{\sigma}^2)$

Naïve: $y_{T+h|T} \sim N(y_T, h\hat{\sigma}^2)$

Seasonal naïve: $y_{T+h|T} \sim N(y_{T+h-m(k+1)}, (k + 1)\hat{\sigma}^2)$

Note that when h=1 and T is large, these all give the same approximate forecast variance

Standard ML metrics are:

$$\text{MAE} = \text{mean}(|e_{T+h}|)$$

$$\text{MSE} = \text{mean}(e_{T+h}^2) \qquad \text{RMSE} = \sqrt{\text{mean}(e_{T+h}^2)}$$

$$\text{MAPE} = 100\text{mean}(|e_{T+h}|/|y_{T+h}|)$$

For time series, it is usually better to normalize these scores.

Example: mean absolute scaled error $MASE : mean(|e_{T+h}|/Q)$. Here Q is a normalizing factor.

For non-seasonal time series

$Q = (T-1)^{-1} \sum_{t=2}^{T} |y_t - y_{t-1}|$

is equivalent to MAE relative to a naïve method.

For seasonal time series

$Q = (T-m)^{-1} \sum_{t=m+1}^{T} |y_t - y_{t-m}|$

is equivalent to MAE relative to a seasonal naïve method

Mean method

The mean model is

$$\hat{y}_{t+1} = c + \varepsilon_t$$

Where $\epsilon_t \sim \mathcal{N}(0, \sigma)$. For this method, we need to estimate $_c$, which is done via empirical mean:

$$\hat{c} = \frac{1}{T} \sum_{t}^{T} y_t$$

Mean method -2

Since the model treats the predicted variable as constant, the prediction error is independent of the step ahead h. The predictive distribution is Gaussian, with variance:

$$Var(\hat{y}_{t+h}) = Var(\hat{y}_t) = Var(\hat{c}) + Var(\varepsilon_t) = \frac{\sigma^2}{T} + \sigma^2 = \left(1 + \frac{1}{T}\right)\sigma^2$$

The first quantity is the variance associated to the estimation of the parameter c, and stems directly from the central limit theorem: the uncertainty in the estimation of the distribution mean through empirical mean is normally distributed with variance ~ (1/T).

Naive method

The naive method assumes that the true data generation process has the following form:

$$y_{t+1} = y_t + \varepsilon_t$$

We can obtain the formula for the h_{th} step ahead prediction as:

$$\begin{aligned}
y_{t+1} &= y_t + \varepsilon_t \\
y_{t+2} &= y_{t+1} + \varepsilon_{t+1} = y_t + \varepsilon_t + \varepsilon_{t+1} \\
&\vdots \\
y_{t+h} &= y_t + \sum_{i=1}^{h} \varepsilon_{t+i},
\end{aligned}$$

Naive method - 2

Since the errors are considered to be zero mean, the predictive variable and its variance are:

$$\hat{y}_{T+h|T} = E\left(y_{T+h|T}\right) = y_T$$
$$\hat{\sigma}_h^2 = V\left(y_{T+h|T}\right) = h\sigma^2$$

Where the last expression is due to the fact that a summation of Gaussian random variables is still a Gaussian with variance equal to the sum of their variances.

Seasonal method

The seasonal naive method assumes the following data generation process
$\hat{y}_{t+h|t} = y_{t+h-m(k+1)}$

This can be rewritten as:

$$
\begin{aligned}
y_{t+1} &= y_{t+1-m} + \varepsilon_{t+1} \\
y_{t+2} &= y_{t+2-m} + \varepsilon_{t+2} \\
&\vdots \\
y_{t+m} &= y_t + \varepsilon_{t+m} \\
y_{t+m+1} &= y_{t+1} + \varepsilon_{t+m+1} = y_{t+1-m} + \varepsilon_{T+1} + \varepsilon_{t+m+1} \\
&\vdots \\
y_{t+2m} &= y_{t+m} + \varepsilon_{t+2m} = y_t + \varepsilon_{t+m} + \varepsilon_{t+2m} \\
y_{t+2m+1} &= y_{t+m+1} + \varepsilon_{t+2m+1} = y_{t+1-m} + \varepsilon_{t+1} + \varepsilon_{t+m+1} + \varepsilon_{t+2m+1} \\
&\vdots
\end{aligned}
$$

Seasonal method - 2

Thus:

$$\hat{y}_{T+h|T} = E\left(y_{T+h|T}\right) = y_{T+h-m(k+1)}$$
$$\hat{\sigma}_h^2 = V\left(y_{T+h|T}\right) = (k+1)\sigma^2$$

For ARCH(1) the raw ACF can look white and both portmanteau tests may not reject, while the ACF and Ljung–Box test of x_t^2 clearly reveal serial dependence. That makes the essential point: these diagnostics test **linear autocorrelation**, not independence or all nonlinear dependence.